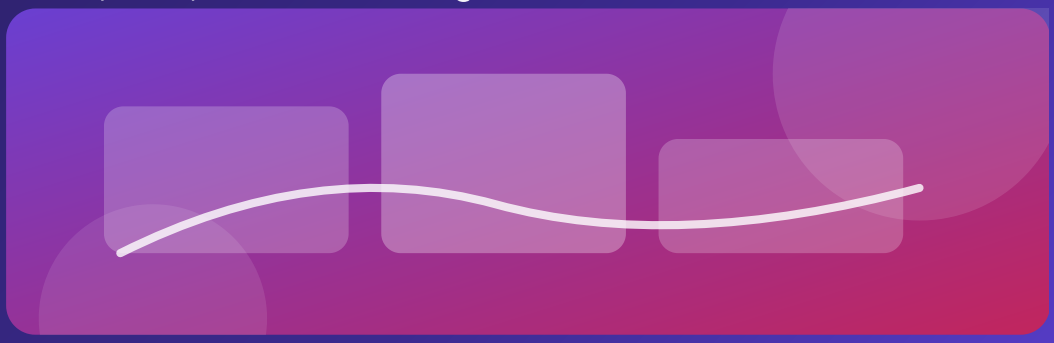
SkillNavora · Product Management Master Handbook

Module 10: LLMs, RAG and AI Foundations

Detailed supplement: how LLMs work in plain English, the PM glossary, the prompt quality ladder, the PREP and CRISP-style briefs, RAG, and a PRD assignment



Written by Varun Kumar, for SkillNavora · Draft

Official site: www.skillnavora.in

Prepared for Skillnavora — Learn. Practise. Implement. Grow.

MODULE 10 · HOW TO USE THIS SUPPLEMENT

Coverage map: what is new and what is already in the module

The main Module 10 handbook covers tokens, embeddings, prompts and RAG at length. This supplement adds what the Week 10 deck teaches as a PM tool: the plain-English LLM model, the glossary, the prompt quality ladder, the PREP brief with a CRISP-style checklist, the tone lever, RAG in practice, and the PRD assignment.

Topic (Week 10 deck)	Handbook section	Status here
LLMs as next-word prediction machines	10.x	Extended: a simple 'fill in the blank' test
Parameters and training	10.x (partial)	New: a worked picture of how parameters change behaviour
Glossary: token, temperature, context window	10.x	Extended: a one-line PM glossary
Tokenisation and embeddings	10.x	Extended: input and output, step by step
Common pitfalls: hallucination, bias, overfitting	10.x (partial)	New: each pitfall with a PM action
Prompt quality ladder	—	New: vague, clear, contextual, strategic
PREP framework	10.x	Extended: purpose, requirements, examples, parameters
Role, context, instruction and specification	10.x (partial)	New: a worked brief with each part
Tone as a parameter	—	New: the same product in four tones
RAG for PMs and the pipeline	10.x	Extended: scenarios and where PMs decide
PRD assignment with a system prompt	—	New: the three-step brief

NOTE ON THE SOURCE

Some deck slides are pictures with no readable text. Examples in the deck are teaching examples. Model names, prices and product features change; check the current vendor documentation before you rely on them.

EXTENDED · HOW LLMS WORK

An LLM is a next-word prediction machine

An LLM is a very large statistical model. It repeatedly predicts the next word in a sequence, based on patterns it learned from a huge body of text. Its output sounds right because it follows learned patterns. It does not check facts by default.



From data to output, in four steps

Parameters are the internal values the model adjusts during training. One value might help the model recognise a language pattern; changing it changes how the model predicts. Billions of parameters define the model's quality.

EXTENDED · PM GLOSSARY

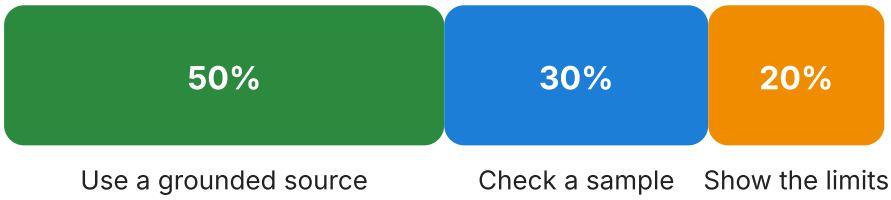
The words you will hear in every AI meeting

Term	Plain meaning	Why a PM cares
Token	A piece of text (a word or part of one) that the model processes	Pricing and limits are often counted in tokens
Parameter	An internal value the model learned	Model size and quality follow from it
Temperature	Controls randomness: lower is more precise, higher is more creative	Pick a setting that fits the task
Context window	How much text the model can use at once	Long documents may not fit
Embedding	A list of numbers that captures the meaning of text	Lets search find similar meaning, not only matching words
Hallucination	A confident answer that is false	Needs a check or a grounded source

NEW · PITFALLS

Hallucination, bias and overfitting: each needs a PM action

Pitfall	What it is	PM action
Hallucination	False confidence in an answer	Ground answers in sources; show citations; test a sample of outputs
Bias	The data is imbalanced, so answers favour some groups	Compare results across user segments; document the risk
Overfitting	The model works on examples it saw and fails on new ones	Test on new, unseen inputs before launch

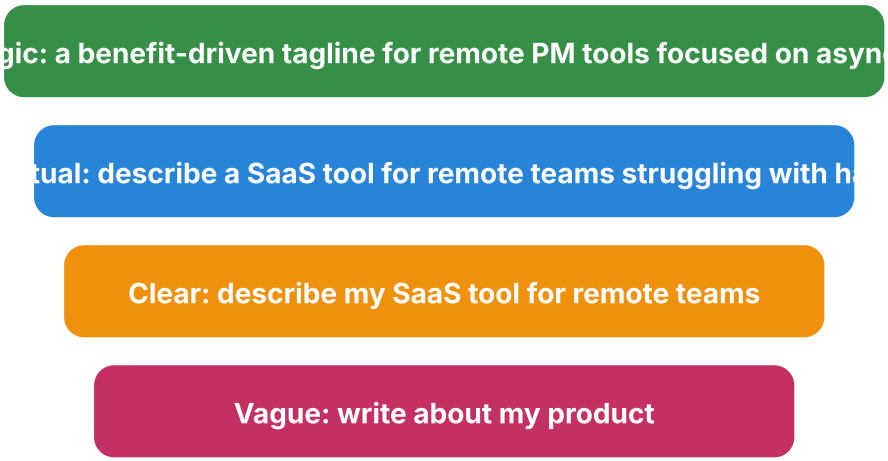


A simple response to hallucination

NEW · PROMPT QUALITY

The prompt quality ladder

Each rung on the ladder adds clarity and strategic focus. The higher you climb, the more useful the answer for your product.



The prompt quality ladder, from vague to strategic

A prompt is a brief, not a question. Give the AI a role, the context, the instruction and the specification.

EXTENDED · THE PREP BRIEF

Purpose, Requirements, Examples and Parameters

Part	What to write	Example
Purpose	The deliverable, specifically	A feature announcement email for a task app
Requirements	Tone, audience, key themes, constraints	Friendly tone, for busy team leads, three key benefits
Examples	One or two samples of 'good'	A sample subject line and an opening sentence
Parameters	Length, format, style	Under 120 words; plain text; no emojis

Role, context, instruction, specification

Part	Example (LinkedIn post for aspiring PMs)
Context	I am a career coach and PM mentor writing short posts for aspiring PMs.
Role	Act as a senior LinkedIn content strategist who writes short, clear posts.
Instruction	Write one post based on my idea. Make it relatable and easy to read.
Specification	Hook under 8 words; short sentences; plain English; no jargon.

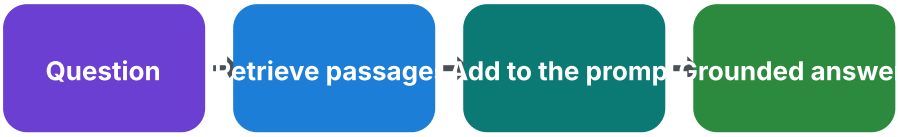
TONE IS A PARAMETER

The same product can speak to different audiences. Test a few tones: inspirational, playful, professional. Keep the one that fits your brand and your audience.

EXTENDED · RAG

Retrieval plus generation gives grounded answers

Retrieval-augmented generation (RAG) first searches your own documents for relevant passages, then asks the model to answer using them. The answer is grounded in your source, which reduces hallucination and keeps answers current.



A RAG pipeline in four steps

Use case	What RAG gives you
Support chatbot	Consistent answers from your help documents
Product research	Faster answers from competitor and user notes
Internal tools	Instant retrieval of company knowledge

BUILD A MINI ASSISTANT (DECK TASK)

Input: your FAQ or documents. Retrieval: a contextual search. Output: a clear answer in a Slack message or a sheet. Design the flow on paper before you build it.

ASSIGNMENT

Design a reusable system prompt, then generate a PRD

The assignment shows that prompt engineering is a form of process design. You design one reusable system prompt, then use it to produce a real deliverable.

1. **Design the system prompt:** give the model a persona (for example, a senior PM with B2B SaaS experience), the exact PRD sections it must produce, and the tone and depth you expect.
2. **Generate the PRD:** use your system prompt to produce a full product requirements document, for a real product or a hypothetical feature.
3. **Prepare the submission:** submit the system prompt and the PRD it generated, as the deck's guidelines require.

SELF-CHECK

Did the prompt name the sections? Did you check every fact in the PRD yourself? Did you note what you changed and why?

Quiz with answers

1. What does an LLM predict?
   Answer: the next token, based on learned patterns.
2. What does a higher temperature do?
   Answer: more randomness and creativity; lower is more precise.
3. What is RAG?
   Answer: retrieve relevant passages from your documents, then generate an answer grounded in them.
4. Which rung of the quality ladder is 'strategic'?
   Answer: the one that adds a specific goal and audience, such as a benefit-driven tagline.
5. What does PREP stand for?
   Answer: Purpose, Requirements, Examples, Parameters.

MODULE 10 · FINAL

Completion checklist

Completion checklist

Done	Item
☐	Read the coverage map
☐	Explained an LLM in one sentence
☐	Wrote a prompt with role, context, instruction and specification
☐	Climbed a prompt from vague to strategic
☐	Sketched a RAG pipeline for one FAQ
☐	Completed the PRD assignment with a system prompt